

Table 4: Example cases from the Callhome testing set where we see big absolute WDER reduction (Δ WDER) with DiarizationLM (deg2ref flavor).

Utterance	Before DiarizationLM	After DiarizationLM
en_6447 (Δ WDER =12.49%)	... <spk:1> i'm <spk:2> going to see if i can talk to the guy that's selling the trailer if i can chew him down a bit uhuh <spk:1> and <spk:2> you know what you just said benedicta is are you living with benedicta <spk:1> yes yes yes <spk:2> you know what i bet she answered the phone ...	... <spk:2> i'm going to see if i can talk to the guy that's selling the trailer if i can chew him down a bit <spk:1> uhuh <spk:2> and you know what you just said benedicta is are you living with benedicta <spk:1> yes yes yes <spk:2> you know what i bet she answered the phone ...
en_6408 (Δ WDER =10.87%)	... <spk:1> uhu <spk:2> so <spk:1> he had big surgery again and he's in a wheelchair oh my <spk:2> and <spk:1> he doesn't want to go to school in a wheelchair uhuh but <spk:2> he might he wants to have tutoring at home but they're still where they lived on 45th street <spk:1> yeah they're there ...	... <spk:2> uhu <spk:1> so he had big surgery again and he's in a wheelchair <spk:2> oh my <spk:1> and he doesn't want to go to school in a wheelchair <spk:2> uhuh <spk:1> but he might he wants to have tutoring at home <spk:2> but they're still where they lived on 45th street <spk:1> yeah they're there ...
en_6298 (Δ WDER =9.95%)	... <spk:1> um hey we're we're confused about you guys address <spk:2> is <spk:1> it 1324 or 13 <spk:2> it's 1 324 <spk:1> excuse me 1324 yes and it's me view is me two words or one word yes it's two words and there's an ln besides ...	... <spk:1> um hey we're we're confused about you guys address is it 1324 or 13 <spk:2> it's 1 324 <spk:1> excuse me 1324 <spk:2> yes <spk:1> and it's me view is me two words or one word <spk:2> yes it's two words and there's an ln besides ...
en_4792 (Δ WDER =9.42%)	... <spk:2> yeah well he was at columbia <spk:1> he was there like five years and they turned him down for tenure then he went somewhere else he he was down in college park maryland yeah and he i think he was only non tenure track down there then supposedly supposed to be back in japan now yeah but you know he's he's probably become an english teacher at some unit yeah i know a guy believe it or not i know a guy from manhattan who was up in sapotto his major he did an mba believe it or not he's he's an english teacher now huh ...	... <spk:2> yeah well he was at columbia <spk:1> he was there like five years and they turned him down for tenure then he went somewhere else he he was down in college park maryland <spk:2> yeah <spk:1> and he i think he was only non tenure track down there then supposedly supposed to be back in japan now <spk:2> yeah <spk:1> but you know he's he's probably become an english teacher at some unit <spk:2> yeah <spk:1> i know a guy believe it or not i know a guy from manhattan who was up in sapotto his major he did an mba believe it or not he's he's an english teacher now <spk:2> huh ...

Table 5: Example cases from the Fisher testing set where zero-shot PaLM 2-S deletes lots of text from the prompt.

Utterance	Before DiarizationLM	After DiarizationLM
fe_03_11252	... <spk:1> oh okay i believe it's a lot wrong with the public schools i don't believe that they're um that they're giving um these kids a sense of um well they're not teaching them what they need to know once they get out of um school you know mhm um what what's happening is that's probably why you got a lot of um a lot of people that's unemployed i think you know they you get a lot from school and they taking a lot of um i guess the economics out of school you know <spk:2> right ...	... <spk:1> oh okay i believe it's a lot wrong with the public schools i don't believe that they're um that they're giving um these kids a sense of um well they're not teaching them what they need to know once they get out of um school you know <spk:2> right ...
fe_03_11224	... <spk:1> so um i think what do you think is an important thing in a relation i think the topic was um what you um what are the most important things in a life partner yeah uh h well what do you think me <spk:2> i would have to say trust and honesty like cuz without that you really don't have nothing to build on you know right yeah ...	... <spk:1> so um i think what do you think is an important thing in a relation <spk:2> i would have to say trust and honesty like cuz without that you really don't have nothing to build on you know right ...

Another research direction would be to compare different LLMs, in different size variants on the speaker diarization task. Specifically, the performance will likely be even better if we finetune larger models such as PaLM 2-M or PaLM 2-L. It would also be interesting to reproduce the experiments with other speaker diarization systems such as EEND [20] or WEEND [34].

Lastly, as PaLM 2 models are multilingual [46], the DiarizationLM framework can naturally apply to speaker diarization tasks in other languages. It would be helpful to evaluate how DiarizationLM performs on speaker diarization datasets in other languages than English.

6 RELATED WORK

6.1 SPEAKER DIARIZATION POST-PROCESSING

In the context of conventional speaker diarization, “post-processing” usually refers to a stage where the clustering results are refined with signals from other sources or systems. An early post-processing approach was known as “resegmentation”, where the Gaussian mixture models (GMMs) are estimated for each speaker with the Baum-Welch algorithm, and a Viterbi algorithm is used to re-annotate the speakers with the GMMs [49]. Later in [50], the authors proposed to use a neural network for resegmentation, with an additional class for non-speech. In [51], the authors proposed DiaCorrect, a method inspired by error correction techniques in ASR. DiaCorrect uses parallel convolutional encoders for the speakers from the initial diarization results and a transformer based decoder to produce corrected diarization results. One major difference in our proposed framework is that we leverage semantic information to refine the diarization results on a word level, while these resegmentation approaches are only based on acoustic information and perform at cluster level.

Another type of post-processing is to combine the outputs of multiple speaker diarization systems, e.g. via majority voting [52], speaker matching [53], or both [54]. More recently in [16], the authors proposed to perform speaker diarization on different temporal scales, and combine their outputs via 1-D convolutional neural networks. In [55], the authors proposed to use end-to-end speaker diarization as a post-processing step for initial speaker diarization results of a clustering-based system.

Our proposed framework is generic such that it can apply to either the results of a single speaker diarization system, or to the combined results of multiple speaker diarization systems.

6.2 SPEAKER DIARIZATION WITH SEMANTIC INFORMATION

Apart from the joint ASR and speaker diarization models discussed in Section 1, researchers have also studied various approaches of integrating semantic information into conventional speaker diarization systems. Some of the benefits of DiarizationLM may also be achieved with non-LLM methods.

The most common approach to leverage semantic information is to use ASR word alignments to refine the voice activity detection and initial segmentation [56]. A variant of this approach is to build a speaker turn detection model and segment by speaker turns [57]. In [58], a Gated Recurrent Units (GRUs) [59] based speaker turn probability estimator is trained on top of word embeddings and speaker embeddings, and the estimated probabilities are combined with the adjacency matrix for spectral clustering. Similarly in [7], an end-to-end trained transformer transducer (T-T) [60] based speaker turn detection model is used to constrain the spectral clustering via Exhaustive and Efficient Constraint Propagation (E2CP).

6.3 SPEAKER DIARIZATION WITH LLM

In [35], the authors proposed Speaker Error Corrector (SEC), which aims to solve the same problem as we stated in Section 1. In [35], word embeddings from the ASR transcript are extracted with a pre-trained Roberta-base LM [61]. Then a separately trained transformer encoder takes the word embeddings and the hypothesis speaker labels as input, and produces the corrected speaker labels. The transformer encoder is trained on both simulated diarization errors and real data. The biggest difference from our proposed framework to [35] is that we directly feed the compact pure textual representation of the ASR and diarization results as part of the prompt to the LLM, and directly finetune the LLM to produce the corrected results in the same compact textual representation. Our DiarizationLM is a “text-in, text-out” system, without relying on internal embedding representations from the LLM.

More recently in [62], the authors proposed to use LLM to predict the speaker probability for the next word, and incorporate this probability into the beam search decoding of speaker diarization. Our proposed framework differs from this work by using a single prompt (or several prompts due to LLM input size limit) to post-process the entire results of the speaker diarization system, instead of word-by-word prompting. Additionally, our proposed framework can be more generally applied to any speaker diarization system, instead of requiring word-level speaker probabilities for beam search decoding.

7 CONCLUSION

In this paper, we demonstrate that large language models (LLM) can be used to post-process speaker diarization results, achieving various goals such as improving the readability of the diarization transcript, and reducing the diarization errors. Specifically, we proposed DiarizationLM, a framework where we use a finetuned LLM to refine the results from off-the-shelf ASR and speaker diarization systems. We introduced three different flavors to build the prompt-completion pairs data for finetuning the LLM. Our experiments on Fisher and Callhome datasets show that a finetuned PaLM 2-S model can drastically reduce the word diarization error rates of typical diarization systems like turn-to-diarize.

REFERENCES

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